

Objective:

To implement and demonstrate Single Inheritance in Java, where the Dog class inherits the eat() method from the Animal class.

Steps for Execution:

Create a Java project and package named javacore.

Create a file named Dog.java and enter the given code.

Define the Animal class with the eat() method.

Create the Dog class by extending Animal.

Compile and run the program.

Observe the output.

Algorithm:

Start the program.

Define the Animal class with the eat() method.

Create the Dog class extending Animal.

Define the bark() method in Dog.

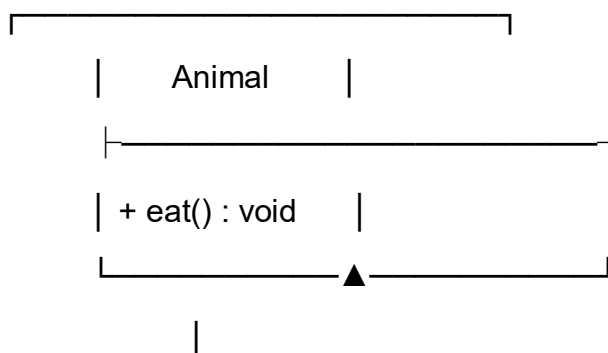
Create a Dog object.

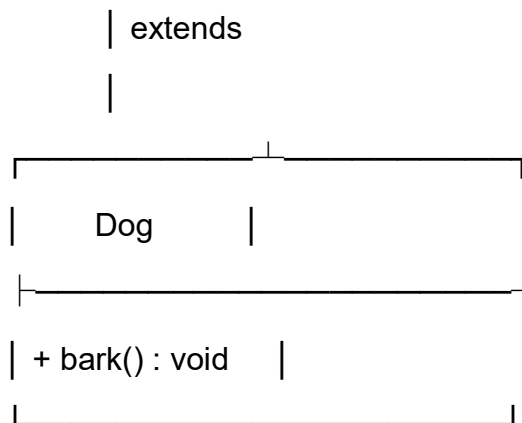
Call eat() and bark().

Display the output.

Stop the program.

UML CLASS DIAGRAM:





Relationship:

- Dog extends Animal
- Dog inherits the eat() method from Animal.

CODE:

```
package javacore;
```

```
class Animal {
```

```
    void eat() {
```

```
        System.out.println("Animal eats");
```

```
    }
```

```
}
```

```
public class Dog extends Animal {
```

```
    void bark() {
```

```
        System.out.println("Dog barks");
```

```
    }
```

```
public static void main(String[] args) {
```

```
    Dog d = new Dog();
```

```
    d.eat();
```

```
    d.bark();
```

```
}
```

```
}
```

OUTPUT:

Animal eats

Dog barks

Result:

The program successfully demonstrates Single Inheritance in Java. The Dog class inherits the eat() method from the Animal class and also contains its own bark() method.